R1.3 Energy from fuels

Combustion reactions

Learning outcomes

By the end of this section you should be able to deduce equations for reactions of combustion, including hydrocarbons and alcohols.

‘Fire has long been perceived as one of, if not the key, human technological innovations, underlying human evolutionary success, including our widespread distribution...’

Katharine MacDonald (2017) (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5489006/)

Without fire, where would we be as a species? Its discovery has given us a way to cook and preserve our food, not to mention providing us with warmth and light. At the root of fire is a chemical reaction known as combustion. Yet, what is combustion and what types of materials can be combusted?

Figure 1. What would have become of humans without the ability to harness combustion reactions?

Credit: Maria Korneeva, Getty Images

A combustion reaction involves two important reactants:

- fuel (a combustible substance)
- oxygen gas.

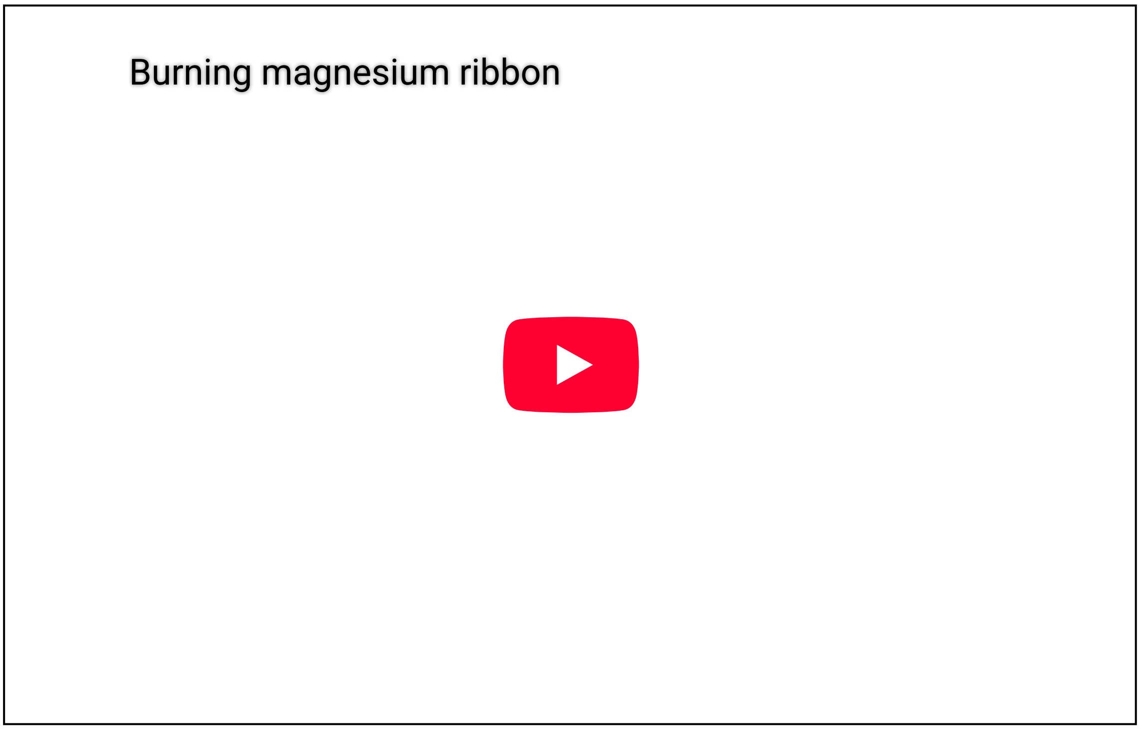
In all cases, the combustible material reacts with oxygen, and produces metallic or non-metallic oxides. These reactions are highly exothermic since the molecules produced are energetically more stable than the reactant fuel and oxygen gas.

Fuels can fall into a variety of categories, from reactive metals and non-metals to organic molecules like hydrocarbons and alcohol. The majority of organic molecules undergo combustion reactions.

Combustion of metals

Nearly all metals react with oxygen to form metallic oxides. Recall from sections S3.1.4 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/group-1-and-17-elements-id-44694/) and S3.1.5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/metallic-and-nonmetallic-continuum-id-44695/), that the most reactive of these metals are the alkali metals followed by the alkaline earth metals. When reactive metals combine with oxygen they can produce tremendous amounts of energy, both in the form of heat and light. This type of reaction is shown below, in the reaction of magnesium metal with oxygen gas.

$$2Mg(s) + O_2(g) \rightarrow 2MgO(s) \quad \Delta H \text{ is negative}$$



Video 1. The combustion of magnesium.

More information for video 1

The interactive video demonstrates the combustion of magnesium ribbon. A strip of magnesium ribbon is clamped in place with a metal stand and ignited using a matchstick flame.
Initially, magnesium is a shiny, silver-grey metal. When heat is applied to one end of the ribbon, the metal reaches its ignition temperature and begins to burn in the presence of atmospheric oxygen.
Upon ignition, the magnesium burns with an extremely bright white flame. As the reaction progresses, it produces a white, powdery ash and a stream of white smoke. This white ash is magnesium oxide (MgO), the product of the reaction between magnesium and oxygen. The aftermath of the reaction leaves behind a noticeable transformation. The once-solid, metallic ribbon has disintegrated into a powdery white substance hanging loosely from the clamp.

Less reactive metals also combine with oxygen. For example, iron, copper, and silver react with oxygen to form their own metallic oxides. We often describe these chemical reactions as rusting or tarnishing. Although the equation showing the reaction of silver and oxygen might look similar to a combustion reaction, the reactions of oxygen with less reactive metals occur at a much slower rate.

$$4Ag(s) + O_2(g) \rightarrow 2Ag_2O(s) \qquad \Delta H \text{ is negative}$$

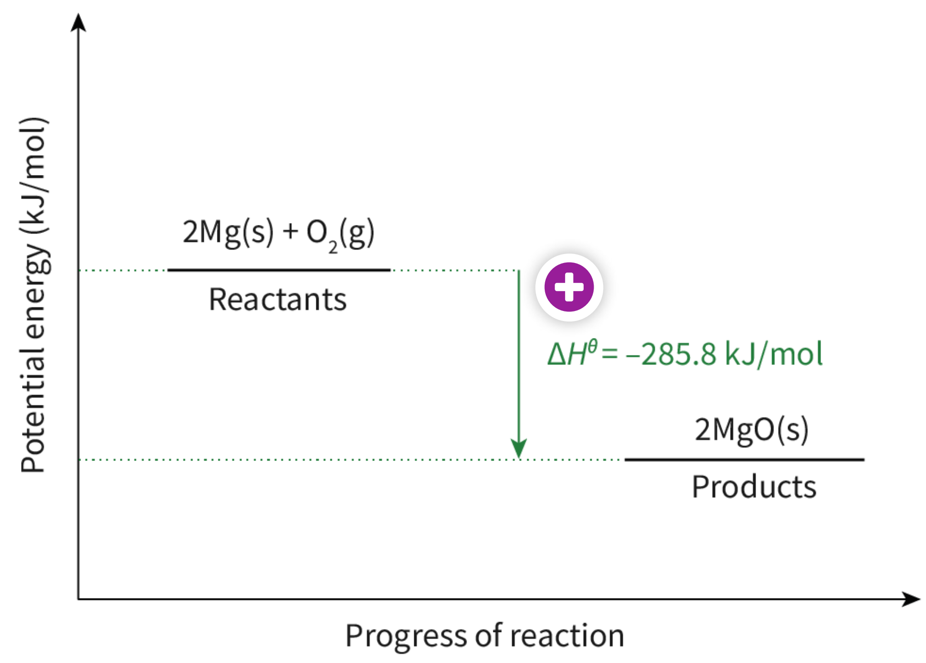
Credit: Brian Caissie, Getty Images

Credit: bridixon, Getty Images

Figure 2. Metal and oxygen reactions.

Making connections

As described in subtopics R1.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43468/) and R1.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43471/), energy is supplied into the system to break the bonds between the reactant atoms. More energy is released to form the bonds within the product molecules than is absorbed to break the bonds in the reactant molecules. As the product molecules form, and become more energetically stable, energy is released from the system to the surroundings.



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Interactive 1. Enthalpy Diagram for the Combustion of Magnesium to Form Magnesium Oxide.

More information for interactive 1

This interactive energy graph illustrates the enthalpy change (ΔH°) during the combustion of magnesium (Mg) to form magnesium oxide (MgO). It visually represents how potential energy changes as the reaction progresses from reactants to products. The graph x-axis represents the progress of the reaction, showing the transition from reactants to products. The y-axis represents the potential energy (kJ/mol). The Reactants: At the beginning of the reaction, magnesium (Mg) and oxygen (O_2) have high energy [$2Mg\,(solid)\ +\ O_2\,(gas)$]. The downward green arrow represents the energy released during the reaction. The enthalpy change (ΔH°) is $-285.8\ kJ/mol$, indicating that energy is lost to the surroundings.
The hotspot represented by a plus sign is given beside the green arrow. By clicking the

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hotspot, it reveals the image of burning magnesium reaction. The image illustrates a bright white flame produced when magnesium metal burns in oxygen. A person is holding magnesium with tongs. The reaction results in the emission of white light and smoke, indicating the formation of magnesium oxide (MgO).
The final energy level is much lower than the reactants. Products: 2MgO (solid).

This graph demonstrates exothermic reactions: The negative enthalpy change (ΔH° equals to minus $-285.8\ \text{kJ/mol}$ means the reaction releases heat. Combustion makes the surroundings warmer. Magnesium burns in the presence of oxygen and results in a bright white flame, producing heat and light energy.

Combustion of non-metals

Conversely, many non-metals react with oxygen to form non-metallic oxides. These oxides do not have fixed ratios, so these combustion reactions can produce several different product oxides for the same non-metal fuel.

Take, for example, the reactions shown below between phosphorus and oxygen gas. Notice that both diphosphorus pentoxide and diphosphorus trioxide can be produced.

$$P_4(s) + 5O_2(g) \rightarrow P_4O_{10}(s) \qquad \Delta H \text{ is negative}$$

$$P_4(s) + 3O_2(g) \rightarrow P_4O_6(s) \qquad \Delta H \text{ is negative}$$

No matter the type of fuel, combustion reactions all involve at least one atom in the combustible material reacting with oxygen. This concept is covered in more detail in subtopic R3.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43486/), but it is still important to realise that oxygen is required for a combustion reaction to take place.

Combustion of organic compounds

In the complete combustion of methane gas, seen below, both the carbon and hydrogen atoms in the reactant molecules experience a gain in oxygen, and are fully oxidised.

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g) \qquad \Delta H \text{ is negative}$$

All organic compounds, especially those that contain only hydrogen and carbon (hydrocarbons), as well as alcohols, easily undergo complete combustion reactions that produce only carbon dioxide and water. Though these reactions always

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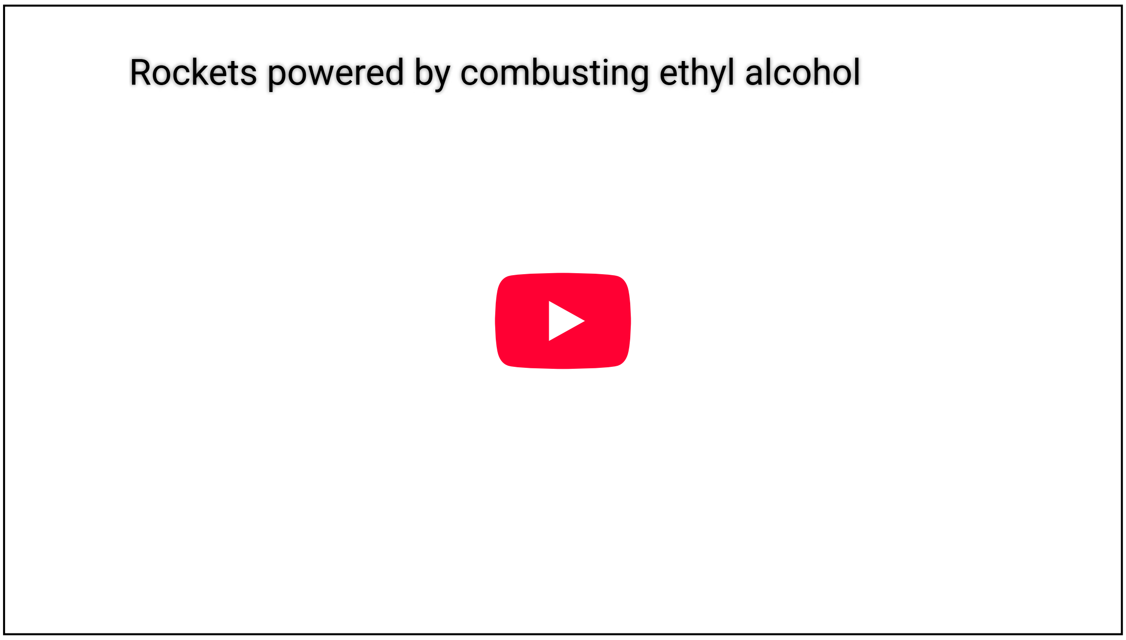
require energy to break the bonds present between the atoms in the fuel and within molecules of oxygen gas, the products are always more energetically stable than the reactants. This means the overall complete combustion reaction is always exothermic!

It is important to realise that organic compounds are flammable no matter the state. It is possible to burn solid carbon (coal), so liquid and gas phase fuels are even more reactive. Always take care when working with fuels, as their vapours easily ignite.

Take for example the complete combustion of ethanol:

$$CH_3CH_2OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(g) \quad \Delta H \text{ is negative}$$



Video 2. Combustion of organic alcohols are highly exothermic reactions.

Making connections

Ethanol can be blended at different ratios with gasoline to create what are known as flex fuels. Ethanol is added to gasoline in many countries to offset the need for fossil fuels. The addition of ethanol to gasoline allows for fewer carbon emissions, as well as a decrease in other types of air pollutants. You will have an opportunity to learn more about these types of fuels in section R1.3.4 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/biofuels-id-45277/) on biofuels.

Activity

- **IB learner profile attribute:**
  - Knowledgeable
  - Thinker
- **Approaches to learning:**
  - Thinking skills — Reflecting on the credibility of results
  - Research skills — Comparing, contrasting and validating information
- **Time required to complete activity:** 20 minutes
- **Activity type:** Pair activity

Instructions

Go on a fuel hunt!

Choose eight different substances — two reactive metals, two non-metals, two hydrocarbons and two alcohols — get creative!

For each substance:

1. Deduce the reaction equation for combustion when heated in oxygen.
2. Research the risks and benefits for the use of each of these substances as a fuel source and for the products formed in each reaction with respect to human health and the environment.
3. Determine which of these fuels would be your preferred choice.